

# Physics 101: Mechanics

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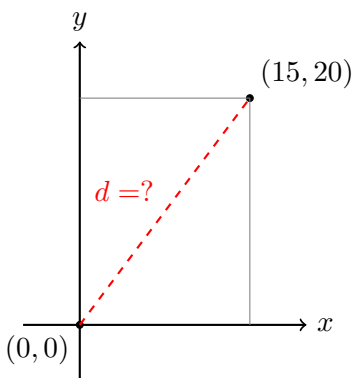
# Contents

# Chapter 1

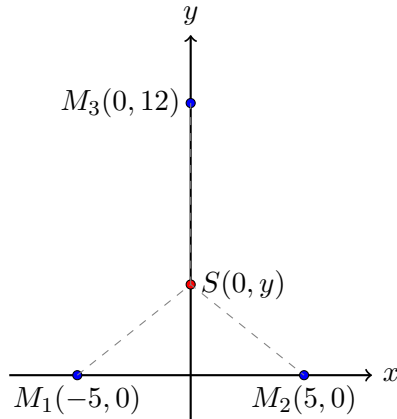
## The Distance Formula using pythagoras

### EXPERIMENTAL LAB: FIELD APPLICATIONS

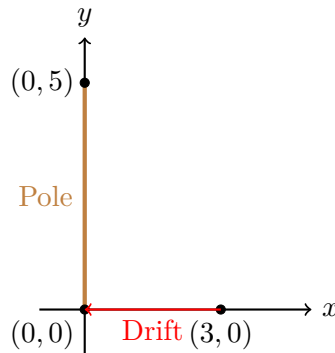
1. **LaserRangefinderCalibration:** A surveyor places a laser at origin  $(0, 0)$  and a reflector at  $(15, 20)$ . The laser reads 24.8 units. Calculate the theoretical distance and find the percentage error.



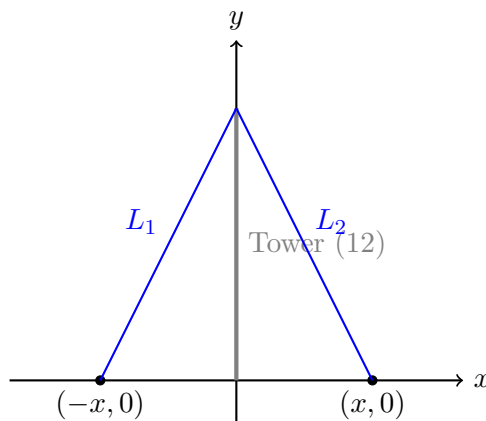
2. **The Acoustic Localization:** Two microphones at  $M_1(-5, 0)$  and  $M_2(5, 0)$  detect sound. A third at  $M_3(0, 12)$  detects it later. Find the source  $S(0, y)$ .



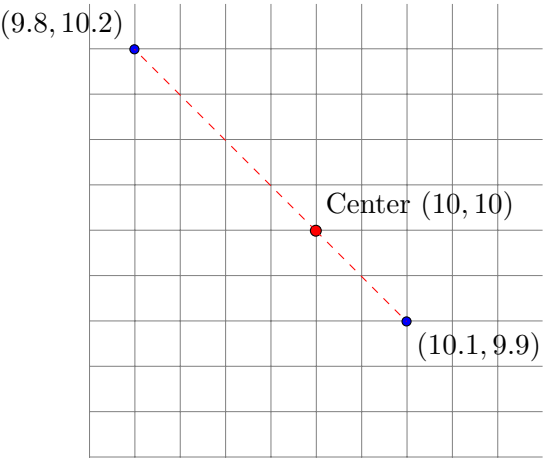
3. **Shadow Tracking:** A pole's top is at  $(0, 5)$  and its shadow is at  $(3, 0)$ . When the sun moves, the shadow moves to  $(0, 0)$ . Calculate the total distance the shadow tip traveled.



4. **Tension Wire Stability:** A tower at  $(0, 12)$  is secured by wires anchored at  $(x, 0)$  and  $(-x, 0)$ . If the total length of both wires is 26, find  $x$ .



5. **GPS Drift Analysis:** Readings at  $(10, 10)$  drift to  $(10.1, 9.9)$  and  $(9.8, 10.2)$ . Find the average distance of these drift points from the center.



## Chapter 2

# Coordinate Geometry

### 2.1 Introduction

Coordinate geometry, also known as analytic geometry, is the study of geometry using a coordinate system. This contrasts with synthetic geometry.

### 2.2 The Cartesian Plane

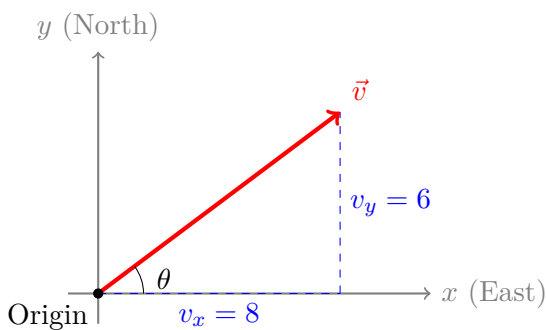
The Cartesian plane is defined by two perpendicular number lines: the x-axis, which is horizontal, and the y-axis, which is vertical.

## Chapter 3

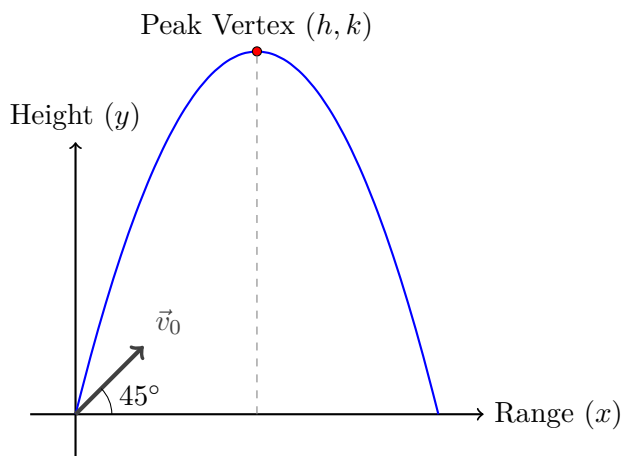
## New Chapter

### EXPERIMENTAL LAB: VECTOR KINEMATICS & DYNAMICS

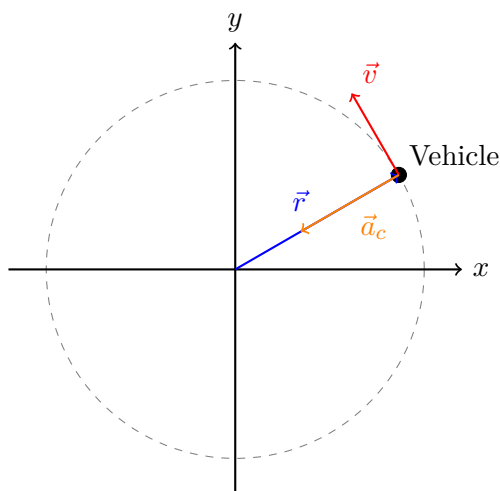
1. **Drone Navigation Vectors:** A quadcopter flies from a launch pad with a velocity vector  $\vec{v} = 8\hat{i} + 6\hat{j}$  meters per second. Calculate its total ground speed  $|\vec{v}|$  and the heading angle  $\theta$  relative to the positive  $x$ -axis.



2. **Projectile Trajectory Analysis:** A ball launcher fires a projectile from ground level at an angle of  $45^\circ$  with an initial speed of 14 m/s. Sketch the parabolic path and compute the peak vertex  $(h, k)$  where vertical velocity  $v_y = 0$ .



3. **Centripetal Force Boundaries:** A tethered test vehicle undergoes uniform circular motion around a central pillar at  $(0, 0)$  with a fixed radius of 5 meters. Show the direction of the centripetal acceleration vector  $\vec{a}_c$  relative to the position vector  $\vec{r}$ .



4. **Static Inclined Plane Equilibrium:** A block weighing 20 N rests on a friction-less ramp inclined at  $30^\circ$ . Draw the complete Free-Body Diagram (FBD) breaking down the gravity vector  $\vec{W}$  into parallel and perpendicular components.



